

Name :- Muskan kumari

Roll no :- 11-25573

Reg no :- 25105142906

Branch :- CSE

1) What are the different protocols you observe at the following layers of protocol stack?

After following the given instruction in the document i am able to capture 16700 packets within 6 mins

i) Application layer

In 16700 packets, found 497 (3.0%) of application protocol which include

- HTTP / JSON → Contain 1 packet i.e less than 0.1%.
- DHCP → Contains 19 packets i.e 0.1%.
- DNS → Contain 2.8% i.e 462 packets.
- HTTP → Contain 16 (0.1%) packets.
- SSDP → Contain 0.3% i.e 55 packets.
- Link - local multicast name resolution → Contain 20% i.e packets in each IPV4 & IPV6.
- Multicast DNS → 9.5% i.e 430 packets in IPV6 & 2.5% i.e 420 packets in IPV4.
- Dynamic host configuration protocol → Contain 0.1% i.e 19 packets in IPV4 & 0.1% i.e 15 packets in IPV6.

ii) Transport layer

- TCP → Contain 38.6% i.e 6393 packets.
- UDP → Contain 26.2% i.e 438 packets in IPV4 & 2.7% i.e 455 packets in IPV6.

iii) Network Layer

- Internet group management protocol $\rightarrow$ Contain 0.6% i.e 95 packet.
- Internet Control message protocol (ICMPv6) $\rightarrow$ contain 1.6% i.e 248 packet.
- IPv4 : Contain 65% i.e 10305 packets
- IPv6 : Contain 4.2% i.e 703 packets

Q.2) What is the total amount of data being received for the following two case?

1) neverssl.com $\rightarrow$ IP Add $\rightarrow$ [34.223.124.45]

Step $\rightarrow$ Filter IP address for specific website used IP.

$$\text{Add} = 34.223.124.40$$

Step II $\rightarrow$ Statics $\rightarrow$ Conversation & then : So found total 200 packet conversion b/w & total amount of data received is if packet ~~length~~ which is 32 KB.

2) Cornell.edu : IP Address $\rightarrow$ [128.253.173.244]

I) found total 72 packet of 29 KB & total packet received from Cornell.edu is 22 KB of total packet 36.

Q.3) Answer the following when you access the site neverssl.com $\rightarrow$

1) What version of HTTP the server running?
 $\rightarrow$ Only HTTP /

2) How many HTTP GET request do you observe?
List down the GET requests.

$\rightarrow$ Total 6 HTTP GET request do you observe?
List from neverssl.com

* HTTP Get request are:-

- i) 579 GET /online HTTP/1.1
- ii) 580 GET /online HTTP/1.1
- iii) 485 GET /favicon.ico HTTP/1.1
- iv) 558 GET /online HTTP/1.1
- v) 559 GET /online HTTP/1.1
- vi) 495 GET /online HTTP/1.1

∴ By using "http.request.method == "GET" and
IP address = 34.223.124.45

3) Identify TCP these way handshake used to establish the connection with ACK packets & write the sequence numbers of each of the three access.

Solve:- Get request

No → 3295 → GET/online HTTP/1.1

Stream no → 101

Number	Information	Sequence no
3198	SYN	0
3295	SYN.ACK	0
3296	ACK	1

Steps which I follow to answer

Step 1 → Get the details of GET HTTP request.

"http.request.method == "GET"

Step 2 → Select any request & check stream index from panel.

Step - 3 → To get TCP handshake details filter using this Command →

"tcp.stream == 101"

Step - 4 → Get the information from info column & non - edum & pannel for sequence no